Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 7 Tutorial Worksheet 2: Placement, Configuration Models and ACLs

Field	Detail
Based on	Week 7 Deck B, Firewall Placement and ACLs lecture (DMZ, Zone-Based Firewall, four placement architectures, Standard/Extended/Named ACLs, ACL placement rules and configuration)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate security technologies), LO5 (configure security technologies)
Format	13 structured questions, no MCQs. Answer in your own words. Several questions ask you to write real ACL syntax.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

The final four questions in the Access Control Lists section ask you to identify the correct ACL parameters for a stated policy, then write the actual configuration. This is deliberate: LO5 is specifically about configuring a security technology to satisfy a requirement, not just describing what the technology can do. Work through the policy carefully before writing any syntax.

The DMZ and Zone-Based Firewall

Q1. Explain what a DMZ is for, and why a compromised DMZ server does not automatically give an attacker access to the internal network. *[LO3]*

Q2. Explain why Zone-Based Firewall is described as a configuration model rather than a new firewall technology. Could a Packet Filtering firewall be configured using zones? Justify your answer. *[LO4]*

Q3. A company configures ZBF with three zones: INSIDE, OUTSIDE, and DMZ. Explain why the OUTSIDE-to-INSIDE zone-pair policy and the INSIDE-to-OUTSIDE zone-pair policy, for the same two zones, would typically not be identical or symmetric. *[LO3, LO4]*

Placement Architectures

Q4. Describe the difference between a single firewall placed behind the border router and a single firewall placed in front of it, and identify which one directly protects the router itself from internet-facing attacks. *[LO4]*

Q5. A financial services company wants the strongest possible defence in depth for its public-facing DMZ, and has the budget to support it. Recommend one of the four placement

architectures covered this week, and justify your choice against at least one alternative. *[LO3, LO4]*

Q6. A small business has a limited budget and a single IT administrator managing everything. Evaluate which placement architecture is most appropriate, referencing both cost and management complexity in your answer. *[LO3, LO4]*

Q7. Explain the specific weakness that the dual-firewall architecture with the router positioned between the two firewalls has, that the dual-firewall architecture with the router positioned before both firewalls does not. *[LO4]*

Access Control Lists

Q8. Explain the implicit deny rule in an ACL, and why forgetting to include an explicit permit statement at the end of an Extended ACL can silently break far more traffic than intended. *[LO5]*

Q9. Explain why Standard ACLs are best placed close to the destination, while Extended ACLs are best placed close to the source. What is it about what each type can actually match that leads to this difference? *[LO4, LO5]*

Q10. A network administrator needs to block only Telnet traffic (port 23) originating from the 10.10.30.0/24 subnet, destined specifically for the server at 172.16.5.10, while allowing all other traffic. Identify the correct ACL type for this policy, and write out the complete ACL statement(s) needed, including where in the network you would apply it and why. *[LO5]*

Q11. A company wants to permit only the management subnet 192.168.50.0/28 to have any access at all through a specific router interface, denying every other source. Identify which ACL type best fits this policy, and write out the complete ACL statement(s) needed. Explain why this type is sufficient here even though it cannot check destination address or port. *[LO5]*

Q12. Convert the following numbered Extended ACL into an equivalent Named ACL called DENY_FTP_ADMIN: access-list 110 deny tcp 192.168.40.0 0.0.0.255 host 10.0.0.50 eq 21, followed by access-list 110 permit ip any any. Show the complete named version, including how you would apply it to an interface. *[LO5]*

Q13. A company's policy states: devices on the 192.168.77.0/24 subnet must not be able to establish new TCP connections to the internal server at 10.1.1.100, but replies to connections that the server itself initiates outward should still be allowed through. Identify the ACL type and the specific keyword needed to implement this policy correctly, and write the ACL statement(s) that would do it. *[LO5]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. This week's lab is direct hands-on practice for the ACL configuration questions above. Actually typing and verifying these commands yourself will cement the theory faster than rereading it.
3. For every parameter-identification question, double-check your answer actually matches the policy as written, not a slightly different policy you assumed. A single word in the scenario, such as new versus reply, can change which keyword the correct answer needs.
4. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5).